

JAY M. FONTICELLA

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EDUCATION AND TRAINING

Case Western Reserve University	Cleveland, OH
Doctoral Candidate of Philosophy in Systems Biology and Bioinformatics	2025 – Present
Harvard University	Cambridge, MA
Postbaccalaureate Fellow in Cancer Biology	2022 – 2025
P.I. Dr. Liron Bar-Peled , CRISPR tiling screens elucidate electron transport chain functional domains.	(23–25)
P.I. Dr. Nika Danial , PUFA metabolism induces a cell state transition in pediatric oncohistone gliomas.	(22–23)
Tufts University	Medford, MA
Bachelor of Science in Biology	2018 – 2022
P.I. Dr. Kelly McLaughlin , Morphological requirements for pre-metamorphic <i>Xenopus laevis</i> craniofacial remodeling.	
GPA: 3.8; Summa Cum Laude; Highest Honors in Thesis	

RESEARCH EXPERIENCE

Massachusetts General Hospital (MGH): The Krantz Family Center for Cancer Research	Charlestown, MA
<i>University of Massachusetts, Boston–Dana-Farber/Harvard Cancer Center (UMB-DF/HCC) U54 Fellowship</i>	July 2023 – July 2025
Ran CRISPR screens with genome-wide cysteine-targeting sgRNA libraries and mitochondrial tiling mutagenesis base-editing libraries, using both proliferation-based and FACS-based readouts. Treated a panel of 40+ cancer cell lines to interrogate how inhibiting mitochondrial translation modulates antineoplastic drug tolerance. To validate hits from the cysteine-targeted CRISPR screens, I induced shRNA-mediated mutations of 2 cysteines on the retromer complex to reveal its new role in ROS sensing and resistance to chemotherapies. I then ran mitochondrial tiling screens to unveil new functional domains in core metabolic enzymes.	
Dana-Farber Cancer Institute (DFCI): Department of Cancer Biology	Boston, MA
<i>Harvard University: The Graduate School of Arts & Sciences NIH-NIGMS PREP Research Scholar Initiative</i>	July 2022 – June 2023
Conducted gas chromatography-mass spectrometry on mouse xenograft biopsies and patient-derived gliomaspheres to profile fatty acid repertoires in oncohistone diffuse midline gliomas, revealing that omega-3 polyunsaturated fatty acids induce a transition from an oligodendrocyte progenitor to an astrocyte-like cell lineage. I next combined bulk-RNA sequencing and biochemical HTRF assays to quantify the changes in transcriptional programs underlying this cell state switch. I then compared ferroptosis vulnerability across the oligodendrocyte progenitor vs. astrocyte cell states comprising these histone 3 mutant brain tumors by flow cytometry and CTG assays. I cultured neurospheres, handled mice, and extracted mouse brains for sectioning.	
National Science Foundation (NSF): International Research Experiences for Students (IRES) Program	Nagoya, Japan
<i>Nagoya University: Institute of Transformative Bio-Molecules Graduate School of Bioagricultural Sciences</i>	May 2022 – July 2022
Screened how ion channel inhibitors perturb real-time transcriptional oscillations in Rat-1– <i>PER2</i> -luciferase fibroblasts. We newly identified 2 ion channels that attenuate <i>PER2</i> transcriptional oscillations upon inhibition. I also trained on tissue extraction in adult <i>Takifugu alboplumbeus</i> and embryonic ribonucleoprotein microinjections to generate knockout zebrafish and medaka.	
National Science Foundation: Louis Stokes Alliance for Minority Participation (LSAMP) Program	Medford, MA
<i>Tufts University: Center for Regenerative & Developmental Biology The McLaughlin Laboratory</i>	September 2019 – May 2022
Completed a senior honors thesis characterizing pharmacologically-induced perturbations in craniofacial cartilage, musculature, and peripheral nerve morphology as well as red blood cell distribution in <i>Xenopus laevis</i> tadpoles during early vs. late pre-metamorphic stages. I applied whole mount immunohistochemistry, geometric morphometric analysis, western blots and immunofluorescence imaging to delineate the molecular requirements for craniofacial tissue remodeling.	
SENS Research Foundation (SRF): Division of Mitochondrial Biology	Mountain View, CA
<i>MitoSENS Engineering Team: The Boominathan Laboratory SRF Summer Scholars Program</i>	May 2021 – August 2021
Optimized a computational pipeline to predict how point mutations in the 13 mtDNA-encoded OXPHOS proteins impacts N-terminal hydrophobicity. I then used PyMOL to assess the consequences on 3D structure and documented the protocol in SOPs.	
Tufts University School of Engineering: Department of Computer Science	Somerville, MA
<i>T-TRIPODS Institute: The Cowen Laboratory NSF HDR Synthetic Coral Project</i>	September 2020 – July 2021
Implemented initial designs of a novel markup language format to validate metadata generated from bulk RNA sequencing of coral holobiont samples. I then coded baseline methods in Python to predict protein function with the MIPS database.	
Johns Hopkins University School of Medicine: Division of Pulmonary and Critical Care Medicine	Baltimore, MD
<i>Bayview Medical Center: NHLBI Summer Internship Program Johns Hopkins Hospital Medical Shadow</i>	May 2019 – July 2019
Measured GSK-induced $[Ca^{2+}]_i$ influxes with live-cell imaging of endothelial cells isolated from rat lungs on temperature-controlled stages. We showed that β -oxidation of palmitate (C16:0) and oleate (C18:1) underlies increased $[Ca^{2+}]_i$ levels and mitochondrial fragmentation in the microvascular endothelial cells of a pulmonary arterial hypertension model. I also rounded with Dr. Stuart Ray during the treatment of HIV patients in the Infectious Disease department.	

Tufts University School of Engineering: The Tissue Engineering Resource Center
NIBIB Biomaterials Core: Intestinal Group | Initiative for Neural Science, Disease & Engineering

Somerville, MA
 October 2018 – March 2020

University of California, Berkeley: The Charles and Louise Travers Department of Political Science
Social Science Matrix: The V&I Laboratory | Human Security, Violence, & Trauma Conference

Berkeley, CA
 May 2020 – May 2021

Categorized how trauma exposure and response correlates to psychosocial, economic, or political outcomes of refugees. I also coordinated panel discussions during the Human, Security, Violence & Trauma conference and submitted a news write-up.

Tufts University School of Arts & Sciences: Department of Psychology

Medford, MA

The Racial Equity & Diversity Laboratory | Discrimination as Trauma Group Research Assistant

September 2020 – December 2020

Completed a literature review on prior measures of brain and cardiovascular activity recorded during vignettes depicting racial discrimination for the background of a 2020 Tufts Springboard grant application, for which we were subsequently funded.

TEACHING EXPERIENCE

Dana-Farber/Harvard Cancer Center: Continuing Umbrella of Research Experiences, Instructor (May – July 2024) **Boston, MA**

Taught high school students about experimental design, oncogenic drivers, and redox biology regulation in cancer research.

Columbia University: YouCanToo Mentorship Program, Forest Hills High School Mentor (August 2020 – May 2021) **Queens, NY**

Advised high school students of color during college applications, wherein my mentees became first-generation college students.

San Francisco State University: Science Coding Immersion Program, Team Lead (June 2020 – July 2020) **San Francisco, CA**

Led daily meetings to train graduate students on using Python scripts to compare CpG mutation frequencies across viruses.

Tufts University: Tufts After School Tutoring Event (TASTE), Student Teacher (September 2018 – April 2019) **Somerville, MA**

Completed interactive STEM activities and grammar lessons with elementary school students in Mystic River public housing.

WRITING EXPERIENCE

Health Across Borders: The Global Modeling Informed Leadership & Connected Community Box

Cape Town, SA

Community Organizing Toolkit: COVID-19 Response Initiative | Grant Writing Fellow

June 2020 – September 2020

Drafted grant applications to fund food delivery services and youth engagement initiatives during the COVID-19 pandemic.

Tufts University: Breakthrough Interdisciplinary Science Magazine

Medford, MA

Undergraduate Academic Research Club: Speaker Series Facilitator | Publicity Director

September 2018 – April 2019

Edited student write-ups for magazine issues and organized speaker series discussing research careers in academia and industry.

HEALTHCARE EXPERIENCE

Timmy Global Health (TGH): La Clínica 32 Volcánes

Quetzaltenango, Guatemala

Assistant Medical Program Coordinator | Medical Scribe-Translator | Tufts TGH Chapter President

December 2018 – May 2021

Recorded patient vitals and co-coordinated clinics with community leaders in Xeabaj, Pujujil, and Llanos del Pinal. I tracked 0–5-year-old child malnutrition progress by comparing head circumference and body dimensions to standard values of the World Health Organization's Anthropometric Calculator. I also scribed and translated medicine intake instructions for patients and doctors in K'iche'-to-English and Spanish-to-English at biannual medical brigades.

Centers for Disease Control and Prevention (CDC): Undergraduate Public Health Scholars (CUPS)

Sacramento, CA

UC Davis Health: The MIND Institute MCHC/RISE-UP | Laboratory on Language Development

May 2021 – July 2021

Examined patterns of mental health symptoms and family well-being scores in caregivers of youth with neurodevelopmental disabilities. We identified caregiver race and child disorder diagnosis as household predictors for family well-being as well as trends in the type of mental health services accessed by caregivers after the onset of the COVID-19 pandemic.

Cambridge Health Alliance: Department of Community Health Improvement

Cambridge, MA

Jonathan M. Tisch College of Civic Life: Community Engagement Intern | Health Data Analyst

June 2020 – August 2020

Engaged community members to identify mental health priorities and other emerging COVID-19 pandemic-related needs. I also analyzed the distribution of health priorities by relevant strata in curated YRBS data. We then presented our findings and community-wide prevention strategies to a MGH Community Advisory Committee during the MassDPH DoN process.

The Scripps Research Translational Institute: The NIH CTSA Scripps Hub Academic Research Core

La Jolla, CA

Scripps Health: Whittier Diabetes Institute | NSF Research Experiences for Undergraduates (REU)

June 2020 – August 2020

Organized COVID-19 perception questionnaires for town hall sessions introducing Moderna COVID-19 vaccination trials.

Robert Wood Johnson Foundation: Summer Health Professions Education Program (SHPEP)

Houston, TX

UTHealth Science Center at Houston: McGovern Medical School | AAMC Pre-Medical Enrichment Scholar

May 2020 – June 2020

Trained at McGovern Human Structure anatomy laboratories in blood collection, wound sutures, and stethoscope techniques. I then presented on the role of healthcare policy on financial protection, holistic nutrition, and overall health outcomes within Black and Indigenous communities. I also completed core academic curriculum on human anatomy and PBL simulations.

Harvard T.H. Chan School of Public Health: Academic Public Health Volunteer Corps (APHVC)

Lawrence, MA

Massachusetts General Hospital (MGH): Community Wellness Collaborative | COVID-19 Contact Tracer

March 2020 – May 2020

Interviewed Spanish-speaking COVID-19 cases regarding ongoing symptoms, quarantine, and traceable contacts. I then connected Spanish-speaking residents to mental health support, prescription delivery systems, and COVID-19 testing sites.

JOURNAL AND NEWS PUBLICATIONS

- 1) Jones, E., **Fonticella, J.M.**, & McLaughlin, K.A., (2025). Identification and characterization of static craniofacial defects in pre-metamorphic *Xenopus laevis* tadpoles. *Journal of Developmental Biology* **13**, 26. doi: 10.3390/jdb13030026
- 2) Zhang, J., Ali, M.Y., Chong, H., Tien, P.-C., Woods, J., Vornbäumen, T., Ordulu, Z., Noble, A., Possemato, A., Harry, S., **Fonticella, J.M.**, ..., Bardeesy, N., Mostoslavsky, R., Pépin, D., Ott, C.J., Liao, B., & Bar-Peled, L., (2025). Oxidation of Retromer Complex controls mitochondrial translation. *Nature* **641**, 1048-1058. doi: 10.1038/s41586-025-08756-y
- 3) Phillip, N., Yun, X., Pi, H., Murray, S., Hill, Z., **Fonticella, J.**, ..., Le, A., Shimoda, L., & Suresh, K., (2024). Fatty acid metabolism promotes TRPV4 activity in lung microvascular endothelial cells in pulmonary arterial hypertension. *American Journal of Physiology-Lung Cellular and Molecular Physiology* **326**, 252-265. doi: 10.1152/ajplung.00199.2023
- 4) Neyazi, S., Gowda, P., Trissal, M., **Fonticella, J.-M.**, Portero, E., Haley-Read, J.-P., Anderson, E., Danial, N., & Filbin, M., (2023). DIPG-28. Identification and targeting of metabolic vulnerabilities in combination with PDGFRA-inhibition in diffuse midline glioma. *Neuro-Oncology* **25**, i19. doi: 10.1093/neuonc/noad073.075
- 5) Shpilker, P., Freeman, J., McKelvie, H., Ashey, J., **Fonticella, J.-M.**, Putnam, H., Greenberg, J., Cowen, L., Couch, A., & Daniels, N., (2022). MEDFORD: A human- and machine-readable metadata markup language. *Database* **2022**, baac065. doi: 10.1093/database/baac065
- 6) Shpilker, P., Freeman, J., McKelvie, H., Ashey, J., **Fonticella, J.-M.**, Putnam, H., Greenberg, J., Cowen, L., Couch, A., & Daniels, N., (2021). METaData Format for Open Reef Data (MEDFORD). *Communications in Computer and Information Science* **1537**, 206-211. doi: 10.1007/978-3-030-98876-0_18

Media Features

1. "Senior Spotlight: Jay M. Fonticella," *Tufts Now | Biomedical Science*, May 2022.
2. "Pamies Scholarship: Shattering Barriers to Pursuing Health Professions Careers," *The SHPEP Connect Newsletter*, June 2022.
3. "2021 Healthline and Prevention Institute Stronger Scholarship Winner: A Chat with Jay-Miguel Fonticella," *Healthline*, September 2021.

POSTERS AND PRESENTATIONS

- 1) **Fonticella, J.M.**, Ali, M.Y., & Bar-Peled, L., (2024). Tiling mutagenesis of the electron transport chain complexes to delineate mitochondrial dependencies in non-small cell lung cancer. UMB-DF/HCC U54 Partnership Scientific Meeting, Boston, MA.
- 2) **Fonticella, J.M.**, Wang, H.-T., Wang, S., Ren, L., Chen, J., Yoshimura, T., & Kon, N., (2022). Elucidating how ion signaling pathways modulate the transcriptional oscillations of core circadian clock genes in Rat-1-PER2-luciferase fibroblasts. Nagoya University Laboratory of Animal Integrative Physiology Progress Seminar, Nagoya, Japan.
- 3) **Fonticella, J.-M.**, Esterman, E., Rao, L., Dixit, B., & Boominathan, A., (2022). A computational predictive analysis on the effects of point mutation induced hydrophobicity changes on import capability and structure of mtDNA-encoded oxidative phosphorylation proteins in the N-terminal regions. The Society for the Advancement of Chicanos/Hispanics and Native Americans in Science (SACNAS) National Diversity in STEM (NDiSTEM) Conference, San Juan, Puerto Rico.
- 4) **Fonticella, J.-M.**, Bennett, R., Zhang, H., & Wang, A., (2021). Weight-bearing C-arm cone-beam computed tomography (CT) imaging of the knee using a dynamic range reducer (DyRaR). Astronaut Scholarship Foundation (ASF) Innovators Gala & Technical Conference, Orlando, FL.
- 5) **Fonticella, J.-M.**, Bullard, L., John-Thurman, A., & Abbeduto, L., (2021). Family wellbeing and mental health status in caregivers of children with neurodevelopmental disabilities during the COVID-19 pandemic. Centers for Disease Control and Prevention (CDC) Undergraduate Public Health Scholars (CUPS) Student Showcase & Exposition, Baltimore, MD.
- 6) **Fonticella, J.-M.**, Jones, E., Lukason, D., Levin, M., & McLaughlin, K., (2020). The characterization of remodeling and non-remodeling craniofacial defects in pre-metamorphic *Xenopus laevis* tadpoles. New England Louis Stokes Alliance for Minority Participation (NE LSAMP) Symposium, Medford, MA.
- 7) **Fonticella, J.M.**, Servinsky, L., Philip, N., Jiang, H., Xun, Y., Huetsch, J., Suresh, K., & Shimoda, L., (2019). Fatty acid oxidation and intracellular calcium levels in rat lung microvascular endothelial cells. Johns Hopkins University School of Medicine Career, Academic, and Research Experiences for Students (C.A.R.E.S) Summer Symposium, Baltimore, MD.

LEADERSHIP AND CIVIC ACTIVITIES

- Tufts University School of Medicine: Sharewood Project**, *Project Coordinator* (December 2019 – May 2021) **Boston, MA**
Led undergraduate efforts to compile a list of primary care providers accepting MassHealth insurance and housing insecurity resources in the Greater Boston area. We then produced a website to advise patients on these community resources in-clinic.
- Tufts Kidney Disease Screening & Awareness Program**, *Community Screening Chair* (May 2019 – May 2021) **Boston, MA**
Coordinated monthly chronic kidney disease screenings to provide physician consults and health education. I trained and managed undergraduate volunteers in blood glucometry, blood pressure readings, and urinalysis stations at open clinics. I also shadowed glomerular disease and CKD consults at the Nephrology Department of Tufts Medical Center.
- Tufts Leonard Carmichael Society Cancer Outreach**, *Hope Lodge Coordinator* (December 2018 – April 2021) **Boston, MA**
Guided undergraduates in serving biannual dinners for cancer patients at the American Cancer Society's Boston Hope Lodge.

HONORS AND FELLOWSHIPS AWARDED

2023-2025	Post-baccalaureate fellowship in Cancer Research, University of Massachusetts, Boston—Dana-Farber/Harvard Cancer Center Partnership to Advance Cancer Health Equity (National Cancer Institute, PACHE trainee)
2022-2023	Post-baccalaureate fellowship in Life Sciences, The Research Scholar Initiative at the Harvard University Graduate School of Arts (National Institutes of Health, PREP trainee)
2022	National Science Foundation (NSF) International Research Experiences for Students (IRES) fellowship
2022	Travel Scholarship; Diversity in STEM Conference, the Society for the Advancement of Chicanos/Hispanics and Native Americans in Science (National Institute of General Medical Sciences, R13 conference grantee)
2022	Presidential Award for Civic Life, Tufts University, Jonathan M. Tisch College of Civic Life
2022	Highest Honors in Thesis, Tufts University, Department of Biology
2022	Summa Cum Laude, Tufts University, College of Liberal Arts
2021	The Astronaut Scholarship (National) - Astronaut Scholarship Foundation
2021	Rubens J. Pamies, MD, Opportunity Fund Scholarship; Summer Health Professions Education Program (SHPEP) National Program Office (NPO)
2021	Healthline and the Prevention Institute, Stronger Scholarship in Health Equity
2019-2021	National Science Foundation (NSF) Louis Stokes Alliance for Minority Participation (LSAMP) fellowship
2018	The College Board, National Hispanic Recognition Program (NHRP) awardee